# Insoo Chung

AI Research Scientist · Models from first label to production

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## EDUCATION

### B.S. Electrical and Computer Engineering · Seoul National University

Expected May 2029

Major (ECE) GPA **4.02 / 4.3** · Cumulative GPA 3.98 / 4.3

Coursework: Random Variables & Random Processes (A+), Circuit Theory & Lab (A+), Programming Methodology (A+), Engineering Mathematics 1–2 (A0, A+), Digital Computer Concept & Practice (A+), Signals & Systems (A0)

## PUBLICATIONS

CVPR 2026 · **DiffBMP: Differentiable Rendering with Bitmap Primitives.** S. Hong*, J. Kim*, D. Kim, **I. Chung**, S. Y. Chun. arXiv:2602.22625 · diffbmp.com

RSNA 2026 · **Native-Space Deep Learning for Reproducible Diffusion MRI Biomarkers of Cognitive Impairment without Spatial Normalization. I. Chung**, W. Jung, J. H. Lee. Scientific presentation, first author.

## EXPERIENCE

### AI Research Scientist · AIRS Medical, Seoul

Jun 2025 – Present

- **Brain DTI (FDA clearance in progress):** solely developed the ML pipeline end to end: native-space tract-core ROI regression for biomarker sampling, age- and sex-normed percentiles, and GPU tractography. On 784 public repeat-scan pairs it was **3.5× more repeatable** than the standard PSMD marker and **~10× faster** (RSNA 2026).
- **Evaluation without ground truth:** built a harness that grades the DTI model on its measured values (rescan stability, agreement with radiologists' group-level conclusions, clinical signal, silent failures) instead of overlap with hand-drawn ROIs. It showed a geometrically more accurate model was less repeatable, which changed the team's optimization target.
- **Brain volumetry robustness:** turned an unexplained out-of-distribution failure into a label-free same-brain benchmark of acquisition shift, then designed physics-informed augmentations that removed most of the drift. Both the benchmark and the fix shipped in the production model.
- **Lumbar spine MRI, 0 → 1:** solely building the product, from defining clinically relevant measurements (vertebrae, discs, canal, herniation, bulging) to the models, bootstrapping segmentation labels from disc-level weak labels with weakly supervised learning and no manual annotation.
- Built foundation-model inference tooling for pseudo-label generation, adopted across my team.

### Research Intern · Intelligent Computational Imaging Lab, Seoul National University

Jun 2025 – Sep 2025

- Built **Dynamic DiffBMP** for video: per-frame warm start, removal of stuck primitives, and freezing of unchanged regions. Cut temporal flicker (tLP) by **53%** across 17 videos with no loss in fidelity.
- Analyzed gradient collision in primitive optimization: per-pixel gradients cancel when summed, so I ranked primitives by per-pixel gradient magnitude for adaptive control, with a CUDA implementation and visualization tooling.
- Contributed structure-aware initialization, multi-loss optimization for structural preservation, and tile-based parallel rendering.

### Corporate Intern · SNU Future Mobility Technology Center, Gyeonggi-do

Jul 2024 – Aug 2024

- Designed autonomous-driving algorithms in ROS and Python, building real-time perception nodes that fuse LiDAR and camera data to improve navigation accuracy.

## SKILLS

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|---|---|
| **ML** | PyTorch, CUDA, MONAI, Weights & Biases; medical imaging (MRI, diffusion MRI), differentiable rendering |
| **Programming** | Python, C/C++, MATLAB, JavaScript, Verilog; ROS |
| **Languages** | English (native-level) |

## CAMPUS INVOLVEMENT

### 16th Cohort Representative · College of Engineering Honor Society "STEM"

Mar 2025 – Present

- Elected to lead 15 cohort members; represented Korea at the Northeast Asia Student Round Table (SRT) 2025 and led career mentoring at the 2025 STEM Vision Fair.

### Engineering Member · SNU Solar Car Team "SNU SOLO"

Mar 2024 – Jun 2024

- Worked on electrical and software integration (battery management, serial and CAN communication) for a Bridgestone World Solar Challenge vehicle.

### Leader · Band Club "PAMOON"

Mar 2023 – Dec 2024